

Predictive Maintenance Advisor: Business Case and Recommendation

Prepared by Harsh Pandey | Data: NASA C-MAPSS turbofan benchmark, subset FD001 (simulated, not client data)
github.com/Harshpandeyiitian04/pdm_advisor | Live dashboard

RECOMMENDATION

Do not roll out the predictive policy yet. Keep the fixed maintenance schedule for now and run a small pilot with plant data before deciding. In this simulation the predictive policy cost 76.6% more than a fixed schedule and let 25 of 100 engines fail, while the fixed schedule had none.

PROBLEM

A maintenance team wants to service engines before they fail, without pulling every engine on a fixed calendar. The question is whether sensor data can tell the team which engines need attention first, and whether acting on those predictions costs less than the alternatives.

WHAT WAS DONE

- Estimated remaining useful life (RUL) for 100 test engines from sensor histories, using rolling mean, standard deviation and change features. RUL was capped at 125 cycles.
- Compared four standard models plus a separate LSTM experiment. Models were chosen on validation error with engines kept apart between training and validation, not on test results.
- Turned predicted RUL into three bands (Critical, Plan soon, Healthy) and simulated the cost of three maintenance policies.

FINDINGS

Model	Validation RMSE	Test RMSE	Test NASA score
Ridge (baseline)	19.01	20.73	1,426
XGBoost (selected)	16.24	18.19 (18.39 after refit)	1,175 (1,087 after refit)
LSTM (separate run)	12.33	14.79	688

Lower is better for all columns. The LSTM scored best but was a single experiment and is not the model behind the API.

Policy (100 engines)	Estimated cost	Simulated failures	Versus fixed schedule
A. Run to failure	\$5,000,000	100	+357%
B. Fixed schedule (139 cycles)	\$1,093,760	0	baseline
C. Predictive maintenance	\$1,931,932	25	+76.6%

Cost assumptions: \$50,000 per failure, \$8,000 per planned visit, \$60 per unused cycle, 15-cycle safety margin. These are illustrative figures, not plant data.

WHY NOT RECOMMEND PREDICTIVE MAINTENANCE YET

- The model sometimes predicts failure too late. Each missed failure costs \$50,000, which outweighs the savings from servicing engines later than the fixed age.
- The fixed schedule does not look at engine condition, but in this simulation it still avoided every failure at lower cost.
- Results cover one operating condition (FD001) and simulated data. There is no field validation or uncertainty estimate.

NEXT STEPS

1. Find out why the model predicts late for the 25 failed engines, and test a larger safety margin.
2. Replace the assumed costs with real downtime, labor and failure costs from a plant.
3. Test on the other C-MAPSS subsets and on real maintenance logs.
4. Add prediction uncertainty and have reliability engineers review the alert thresholds.
5. If the revised numbers beat the fixed schedule, pilot on a small group of engines before any wider rollout.